Data Science Lab in Medicine - Big Data in Health Care

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Lab 3 - Assessment of the performance of a risk predictor

We will work on a dataset from an medical register containing data about patients who received a heart trasnplantation. Data include all patients tranplanted in the period 01/01/2012 - 31/12/2017 (variable "date.htx"). We are interested in developing a risk score to predict death within 1 year after transplant and to assess its prefromance. Time since tranplant is recorded in years. The "event" variable is either 1 (death) or 0 (administrative censoring). The available prognostic factors are: gender, age, bmi category, status of the patient at transplant (1=critical condition, 2A=severe condition, 2B=stable condition), creatinine and azotemia levels at tranpslant.

Read the data and load the packages "survival", "pROC", "riskRegression" and “dcurves”.

Question 1.

Estimate the overall Kaplan-Meier survival function and compute the estimated risk of death within 1 year after transplant

Question 2.

Estimate the overall censoring function and check the number of subject censored within 1 year after transplant

Question 3.

Exclude patients with a potential follow-up < 1 year (i.e. patients transplanted from 01/01/2017 onwards). This will unbiasedly produce a sample where all alive patients have been observed for at least 1 year.

Question 4.

Working on the sample created in 3., estimate the overall Kaplan-Meier survival function and compute the estimated risk of death within 1 year after transplant.

Question 5.

Working on the sample created in 3., fit two Cox PH models: the first including as predictors "gender", "age", "bmi.cat" and "status"; the second including also "crea" and "azot". For each patient, calculate the predicted risk of death within one year according to both models.

Question 6.

To check the model calibration, create the calibration plot for predicted risks of both models.

Question 7.

Compare the calibration of the two risk predictors also by calculating the Brier Score.

Question 8.

Check the discrimination performance of the two risk predictors using ROC curve and AUC index. Find also the optimal cut-off value according to the Youden index.

Question 9.

Working on the original dataset (N=805 patients) compute the Brier score (adjusted for censoring) and the AUC index. Perform internal validation using bootstrap to correct for the over-optimism.

Question 10.

Calculate and plot the expected Net Benefit according to the risk threshold for both risk predictors.